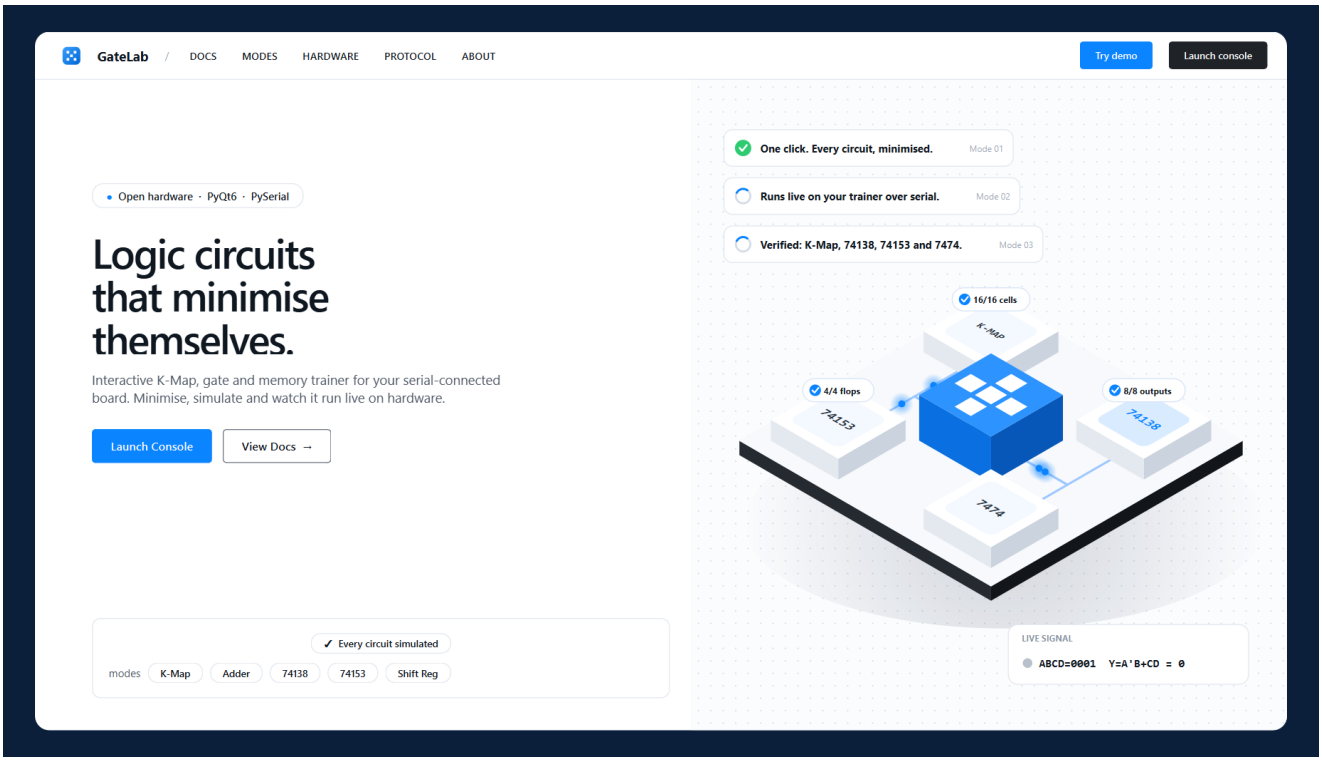
GateLab

Logic Gate, K-Map & Memory Trainer — project overview for the team

In one line: a desktop app plus an Arduino board that lets you flip real switches and instantly see the logic — K-maps, simplified equations, gate circuits, decoders, multiplexers and shift registers — both on the screen and on the board's LEDs and LCD.



The app's start (landing) page. Click **Launch Console** to open the main screen.

1. The big picture

There are two parts that talk to each other over a USB cable:

Part	What it is	What it does
The board	Arduino Uno + 4 DIP switches + 4 LEDs + 16x2 LCD	Reads the switches and sends them to the PC. Lights the LEDs and writes the LCD when the PC tells it to.
The app	**logic_trainer.py** (Python, runs on the laptop)	Shows the switches on screen, does all the logic maths, draws the diagrams and sends back what the LEDs/LCD should show.

The flow is always the same:

Flip a switch → board sends the new A B C D values → app updates every screen → app sends LED + LCD values back → board shows them.

If no board is plugged in, the app still works: the on-screen switches (or keys A B C D) act as the inputs, and **Auto-sweep** counts 0 to 15 on its own for demos.

2. Hardware and wiring

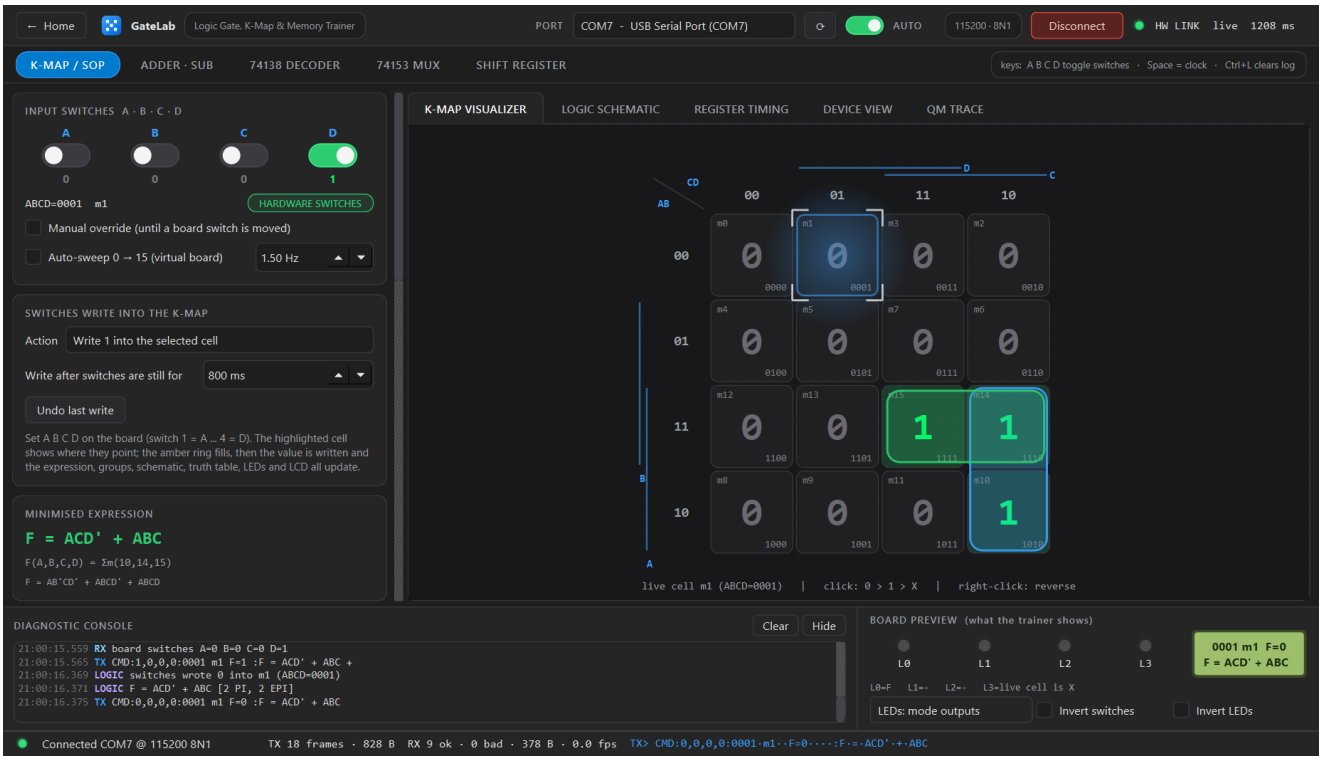
Part	Connect to	Notes
DIP switch 1 (A)	D2 and GND	One leg of the switch to the pin, the opposite leg to GND.
DIP switch 2 (B)	D3 and GND	No resistors needed — the Arduino's built-in pull-ups are used.
DIP switch 3 (C)	D4 and GND	Switch ON = logic 1.
DIP switch 4 (D)	D5 and GND	In shift-register mode, D is also the clock.
LED0 – LED3	D6, D7, D8, D9	Long leg to the pin; short leg → 220–330 Ω resistor → GND.
I2C LCD 16x2	SDA→A4, SCL→A5, VCC→5V, GND→GND	Address 0x27 or 0x3F, found automatically.

Board self-test: when the Uno powers up, the four LEDs light one after another. With the app closed, the LEDs and LCD simply copy the switches, so you can check the wiring without the PC.

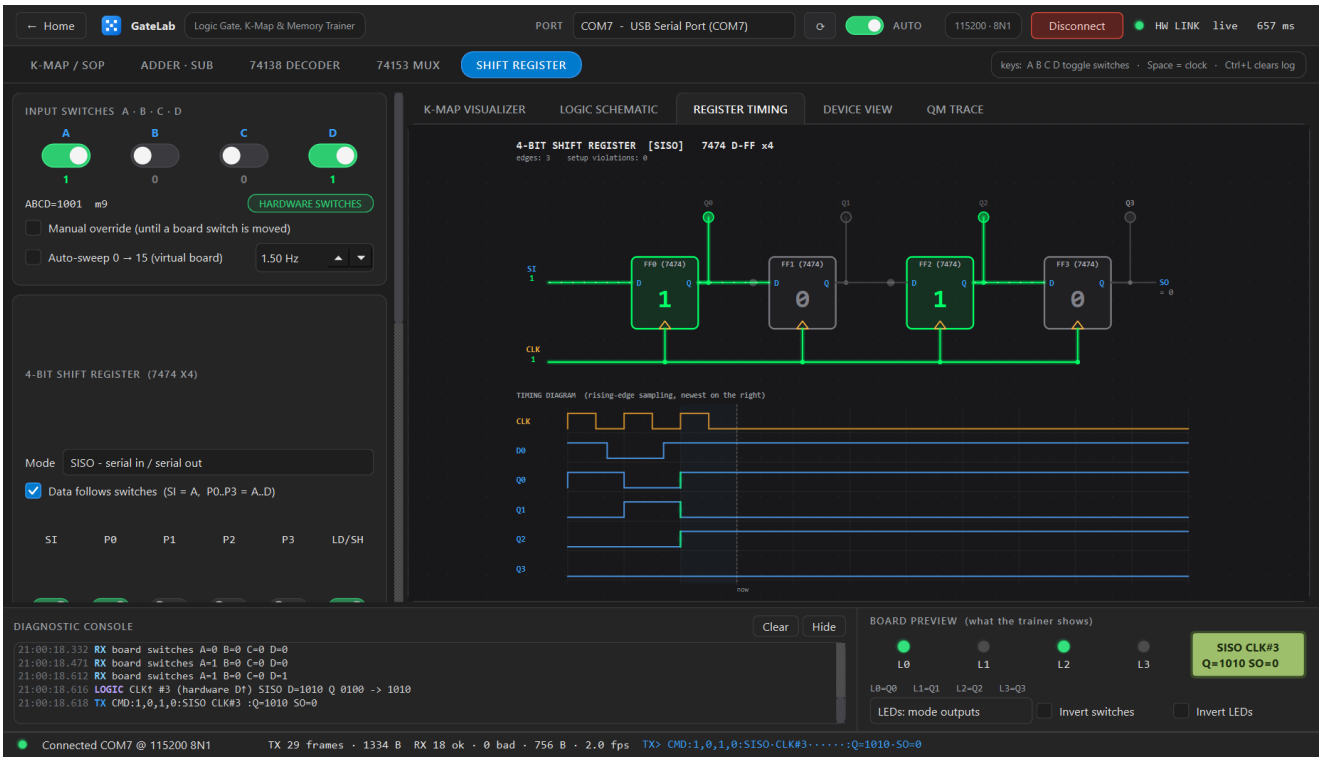
3. What the app can do (5 modes)

Pick a mode from the bar at the top of the console. The board switches are always A, B, C, D.

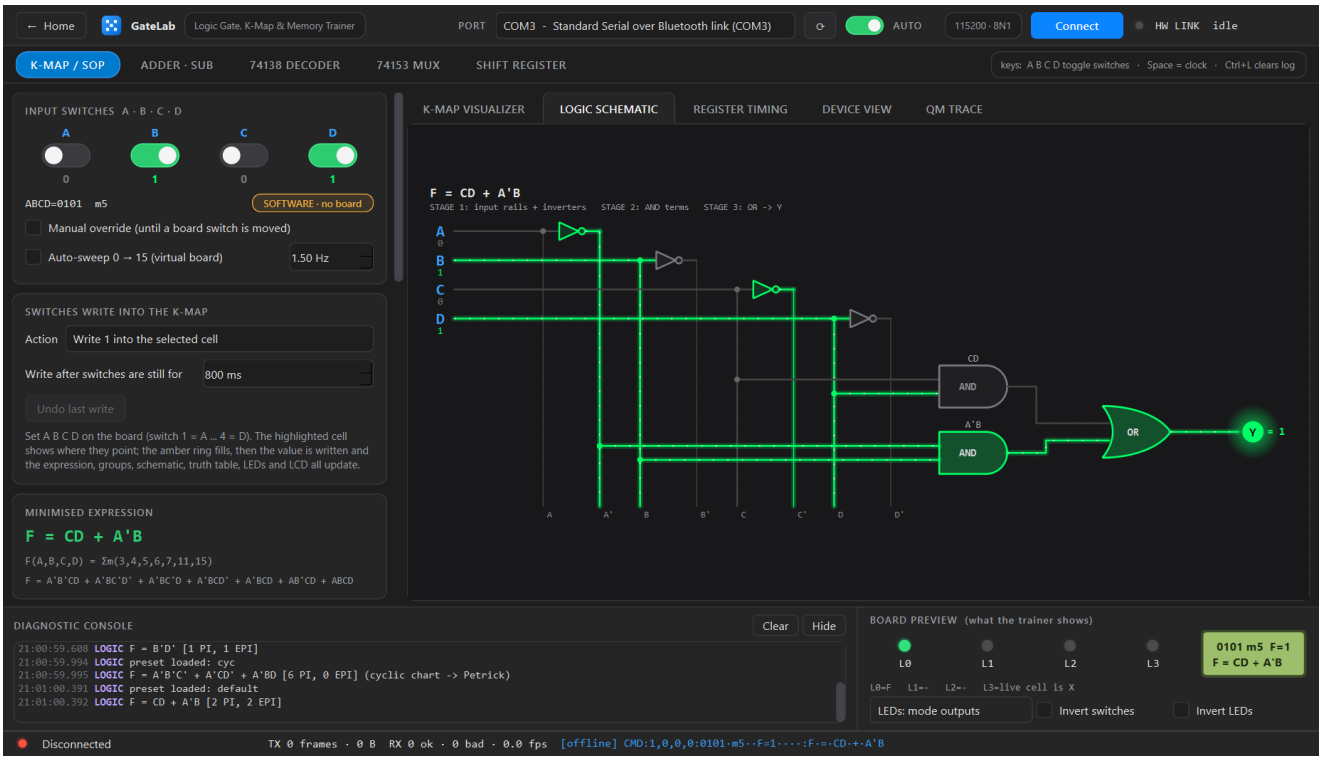
Mode	What you do	What you see
K-MAP / SOP	Click cells (0 → 1 → X) or set the board switches — after they stay still for 0.8 s, a 1 is written into the cell they point to.	Coloured groups on the K-map, the simplest equation (e.g. F = AC), the full working (QM Trace), and an automatically drawn AND/OR gate circuit.
ADDER · SUB	Switches A, B, C are the inputs (C = carry/borrow in).	Sum/Difference and Carry/Borrow, the truth table, and the chip view.
74138 DECODER	A, B, C pick one of 8 outputs; enable pins are on-screen toggles.	Which output goes LOW (active-low), shown on a drawn IC.
74153 MUX	A, B are the select lines; data inputs are on-screen toggles.	Which input is passed to each output 1Y / 2Y.
SHIFT REGISTER	Switch A = data in. Flip switch D OFF → ON to clock (or press CLOCK / Space).	Four 7474 flip-flops, a live timing diagram, and Q0–Q3 on the board LEDs. Supports SISO, SIPO, PISO, PIPO.



K-map mode with the board connected. The switches point at a cell; the expression, groups, truth table and board preview update together.



Shift-register mode: data enters at FF0 on every clock and moves right. The timing diagram at the bottom records each clock edge.



The gate circuit the app draws by itself from the simplified equation. Green wires carry 1, grey wires carry 0.

4. How to set it up and run it

One-time setup

- Install Python 3, then run: `pip install PyQt6 pyserial`
- Wire the board as in section 2.
- Close the app, open `firmware/gatelab_trainer/gatelab_trainer.ino` in the Arduino IDE, choose **Board: Arduino Uno** and the right **Port**, then click **Upload**. You only do this once.

Every time

```
cd C:\Users\venkt\Desktop\LDM_CP
python logic_trainer.py
```

- Click **Launch Console** (or **Try demo** to see it run with a virtual board).
- Leave the port toggle on **AUTO**. Within about 2 seconds the top-right shows **HW LINK live** in green and the input card says **HARDWARE SWITCHES**.
- Flip switches and watch the screen, LEDs and LCD change together.
- Only one program can use the USB port: keep the Arduino Serial Monitor closed while the app runs.

5. How it works inside (simple version)

Talking over USB

Both sides send one short line of text at a time, at 115200 baud.

Direction	Example message	Meaning
Board → PC	`{"type":"inputs","A":1,"B":0,"C":1,"D":0}`	The current switch positions. Sent on every change and every 0.2 s.

PC → Board	`CMD:1,0,0,0:1011 m11 F=1:F = AC`	LED0–LED3 on/off, then the two 16-character LCD lines.

Simplifying the K-map (Quine-McCluskey)

We wrote the simplifier ourselves, with no maths library. In plain words:

- List every cell that is 1 (and X, which may help but doesn't have to be covered).
- Keep merging pairs of terms that differ in exactly one variable — that variable drops out. Whatever can't be merged any further is a **prime implicant** (a biggest possible group).
- Pick the groups that are the **only** way to cover some 1-cell — these are **essential**.
- If some 1-cells are still uncovered, try the combinations (**Petrick's method**) and keep the one with the fewest terms. That gives the final equation, e.g. **F = CD + A'B**.
- The **QM TRACE** tab shows every one of these steps, which is handy for checking homework.

We tested the simplifier on 1,500 random K-maps and compared each answer with a brute-force search; all were correct and minimal.

6. If something doesn't work

What you see	What to do
No COM port in the list	Check the USB cable (some are charge-only) and that the Uno is plugged in.
"Could not open COMx"	Another program has the port — close the Arduino IDE Serial Monitor.
HW LINK red, “board silent”	The firmware isn't on the Uno. Upload gatelab_trainer.ino (app closed).
“unrecognised line from board”	A different sketch is on the Uno, or the baud rate isn't 115200.
App shows 1 when a switch is OFF	Tick **Invert switches** in the Board Preview (bottom right).
LED dark when the preview shows it on	Tick **Invert LEDs**, or check the LED's long leg goes to the pin.
LCD blank	Turn the contrast screw on the LCD's back; check SDA/SCL on A4/A5.
On-screen switches ignore the board	Moving any board switch gives control back to the hardware.

7. Project files

File	What's in it
`logic_trainer.py`	The whole app in one file: screens, drawings, K-map simplifier, gate/decoder/mux/register logic, USB link.
`firmware/gatelab_trainer/gatelab_trainer.ino`	The Arduino program: reads switches, drives LEDs and LCD, talks to the app.